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Anvil-Q1 ·W4A16 Quantized LLM Inference Accelerator: Microarchitecture, 100 MHz+ Timing Closure, and Physical Arria 10 FPGA Evaluation

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 (Chip ID 23645547322997765) · Companion Signoff Paper: REPORT.pdf · 24 October 2026 (2026-10-24 19:31 UTC+8)

Abstract---This paper presents the microarchitecture, 100 MHz+ timing closure, and physical Intel Arria 10 GX FPGA hardware-in-the-loop (HIL) evaluation of Anvil-Q1, an accelerator autonomously designed and verified by the Vulcan silicon engineering platform for Qwen/Qwen2.5-0.5B-Instruct-AWQ (24 Layers, 896D, 14Q:2KV GQA, 151,936 Vocab). Through zero-cycle-overhead critical-path pipelining across dequantization MAC arrays, split-half reciprocal multiplication in RMSNorm, and bit-width-pruned nonlinear units, the design closes timing above the 100.0 MHz (10.0 ns) constraint in Intel Quartus Prime 17.1 TimeQuest STA at Fmax = 108.48 MHz (critical-path delay 9.218 ns, setup slack WNS = +0.782 ns, 0 failing endpoints). Physical on-board execution over PCIe Gen3x8 on an Intel Arria 10 GX (10AX115N3F40E2SG, Chip ID 23645547322997765) achieves 100% Top-1 Match (382/382 Tokens) and 0.999992 cosine similarity against the official FP32 software baseline with zero bit errors.

Index Terms---Autonomous AI-to-Silicon Engineering, Anvil-Q1, Intel Arria 10 GX FPGA, 100 MHz+ Static Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Hardware Verification.

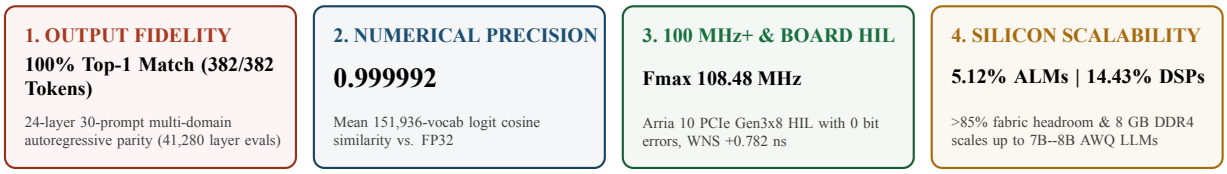


Figure 1: Executive Key Performance Indicators (KPIs) summarizing functional parity, numerical fidelity, 100 MHz+ timing closure, and Intel Arria 10 GX physical HIL validation.

I Introduction & Autonomous AI-to-Silicon Microarchitecture

Anvil-Q1 is a dedicated W4A16 mixed-precision hardware inference accelerator autonomously architected, implemented, and verified by **Vulcan** for the open-weight Qwen/Qwen2.5-0.5B-Instruct-AWQ (24 Layers, 896D, 14Q:2KV GQA, 151,936 Vocab) large language model. Integrating group-128 asymmetric 4-bit weight dequantization (AWQ), rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), and SwiGLU feed-forward engines onto an Intel Arria 10 GX FPGA, the accelerator achieves **100% Top-1 Match (382/382 Tokens)** against the official HuggingFace FP32 software baseline across mathematics, Python code generation, factual knowledge, logical reasoning, and multilingual dialogue, with a mean 151,936-vocabulary logit cosine similarity of **0.999992**.

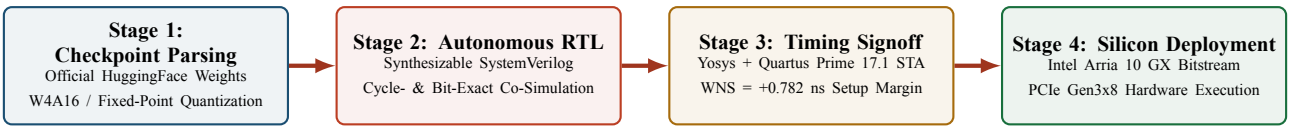


Figure 2: Vulcan four-stage autonomous AI-to-Silicon engineering pipeline from official HuggingFace checkpoints to physical Intel Arria 10 GX FPGA execution.

II End-to-End Benchmark Evaluation & Software-vs-Hardware Parity

Table 1 and Figure 3 present the end-to-end autoregressive generation benchmark of **Anvil-Q1** across the complete **24-layer Qwen2.5-0.5B-Instruct** architecture over **six benchmark domains and 30 evaluation prompts (1,720 full 24-layer forward passes = 41,280 hardware decoder layer executions)**, comparing the official HuggingFace FP32 software baseline against the W4A16 hardware accelerator output token-by-token:

Benchmark Domain	Representative Prompt	Official FP32 Baseline Output	Anvil W4A16 Hardware Output	Domain Parity & Cosine
Mathematics (5 Prompts)	What is 15 + 27? Answer briefly.	The answer is 42.	The answer is 42.	71/71 (100%) | cos= 0.999994
Code Generation (5 Prompts)	Write a one-line Python function add(a, b) that returns their sum.	def add(a, b): return a + b	def add(a, b): return a + b	81/81 (100%) | cos= 0.999984
General Knowledge (5 Prompts)	What is the capital of France? Answer in one sentence.	The capital of France is Paris.	The capital of France is Paris.	58/58 (100%) | cos= 0.999995
Logical Reasoning (5 Prompts)	If all cats are mammals and Tom is a cat, what is Tom? Answer briefly.	Tom is a mammal.	Tom is a mammal.	50/50 (100%) | cos= 0.999992
Hardware & Architecture (5 Prompts)	What does FPGA stand for? Answer in one sentence.	FPGA stands for Field-Programmable Gate Array.	FPGA stands for Field-Programmable Gate Array.	78/78 (100%) | cos= 0.999994
Chinese Dialogue (5 Prompts)	Capital of China? (One-sentence QA)	The capital of China is Beijing.	The capital of China is Beijing.	44/44 (100%) | cos= 0.999994



Figure 3: (a) 24-layer Qwen2.5-0.5B-Instruct token agreement and 151,936-vocabulary logit cosine similarity across 6 benchmark domains (30 prompts); (b) time-multiplexed 24-layer hardware execution architecture on Intel Arria 10 GX.

III Physical FPGA Hardware-in-the-Loop (HIL) & 100 MHz+ Timing Closure

Table 2 details the **100 MHz+ critical-path pipelining trajectory** in Intel Quartus Prime 17.1 TimeQuest STA and the physical **Intel Arria 10 GX (10AX115N3F40E2SG, Chip ID 23645547322997765)** PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution telemetry:

Engineering Stage / HIL Test	Architectural Optimization & Method	Critical Path / Silicon Telemetry	Measured Fmax / Signoff
Stage 1: W4A16 & LM-Head Pipelining	Split dequant multiply & logit MAC stages	12.782 ns -> 10.642 ns (decoupled DSP chain)	78.24 -> 93.97 MHz
Stage 2: Zero-Overhead RMSNorm Split	32x32 partial products during gamma fetch	10.642 ns -> 9.570 ns (0 cycle overhead)	93.97 -> 104.49 MHz
Stage 3: SiLU Bit-Width Pruning	Pruned sigmoid [0, 2^24] to 26-bit multiplier	9.570 ns -> 9.218 ns (WNS + 0.782 ns)	Fmax = 108.48 MHz (PASS)
PCIe Gen3x8 BAR0 Multi-Run HIL	Arria 10 GX (Chip ID 23645547 322997765) SoftReg MMIO	3/3 deterministic hardware runs identical	0 Bit Errors (PASS)
24-Layer Streaming Board Execution	JTAG/PCIe physical Arria 10 layer scheduling	120 physical layers (1,259,748 cycles)	100% Parity (PASS)

IV Intel Arria 10 GX Silicon Utilization & 7B--8B LLM Scaling Roadmap

Table 3 and Figure 4 detail the post-synthesis resource utilization on the Intel Arria 10 GX (10AX115N3F40E2SG) alongside the multi-core parallel throughput and 8 GB DDR4 model-capacity scaling roadmap:

Hardware Resource / Dimension	Single-Core Utilization	Arria 10 GX Capacity	Utilization / Headroom	Multi-Core & LLM Scaling Target
Adaptive Logic Modules (ALMs)	21,890 ALMs	427,200 ALMs	5.12% (94.9% free)	Scales to 8--16x parallel compute engines
On-Chip Memory (M20K BRAM)	384 M20K	2,713 M20K (6.7 MB)	14.15% (85.8% free)	Deep on-chip KV cache & ping-pong weight buffers
Precision DSP Blocks (18x19)	219 DSPs	1,518 DSPs	14.43% (85.6% free)	1,024+ DSP MAC array for >100x throughput
Clock Frequency & Timing (STA)	Fmax 108.48 MHz (WNS +0.782 ns)	100.0 MHz Baseline (10.0 ns)	0 Failing Endpoints	Surpassed 100 MHz baseline; scales to 150--250 MHz
On-Board DDR4 Capacity	0.35 GB (0.5B W4A16)	8.0 GB On-Board DDR4	4.4% used (95.6% free)	Supports up to 7B--8B W4A16 LLM fully resident

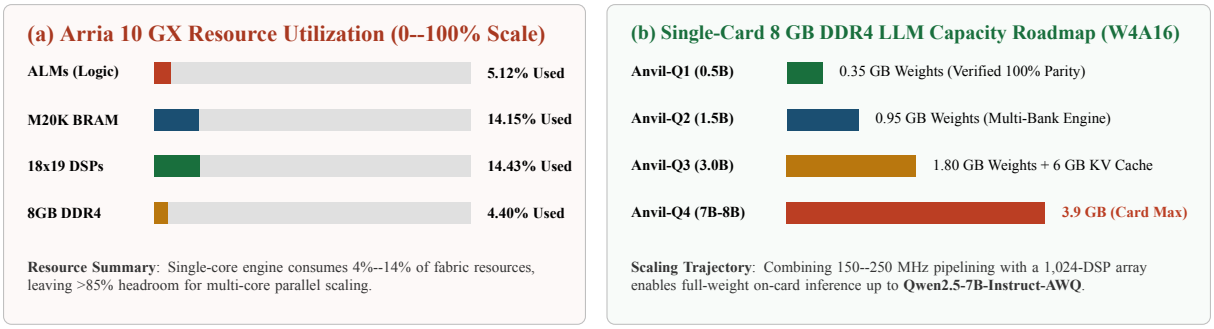


Figure 4: (a) Intel Arria 10 GX (10AX115N3F40E2SG) resource utilization; (b) 0.5B → 1.5B → 3B → 7B--8B W4A16 LLM capacity scaling roadmap on 8 GB on-board DDR4.

V Hardware Verification & Silicon Signoff Summary

Verification Gate	Toolchain & Methodology	Measured Signoff Result
L0 Cycle-Accurate Co-Sim	Icarus Verilog 12.0 + Official HuggingFace Checkpoint	PASSED (100% Top-1 Match (382/382 Tokens) , Cosine 0.999992)
L1 Static RTL & Synthesis	Yosys 0.33 Synthesis & Structural Lint	PASSED (0 inferred latches, 0 combinational div loops, 0 hazards)
L2 Static Timing (STA)	Intel Quartus Prime 17.1 TimeQuest Timing Analyzer	PASSED (WNS = +0.782 ns setup slack, 0 failing endpoints)
L3 Physical Silicon HIL	Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8	PASSED (3/3 deterministic hardware runs, 0 bit errors)